

Orbital state and elements

Thursday, September 3, 2026 10:40 AM

Go from 2D orbit in space to

- 3D
- Orbit in time

Orbital elements and state vector

The state of a system

Lecture 3

9/8/2026

Conserved quantities

- Energy (1)
- Angular momentum (3)
- Eccentricity vector (3)

These are constant, so true anomaly can't be extracted

7 conserved

$3 + 3 + 1 = 5$?

$$\vec{h} \perp \vec{e} \Rightarrow$$

One of these, only two components are required

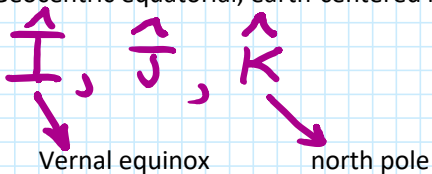
Energy is redundant, it can be calculated from angular momentum and eccentricity

Orbit in three dimensions: embed the orbital plane in 3D space

Three orbital elements from the plane (physics)

Three orbital elements from the 3D embedding (geometry)

Geocentric equatorial, earth-centered inertial (ECI) frame

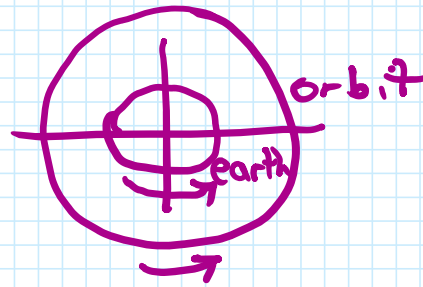
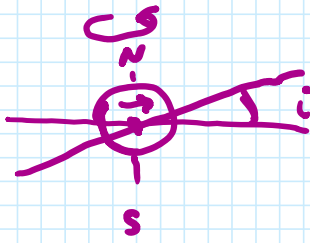


"inertial" -> "non-rotating", i.e., not rotating with the central body

3-1-3 Euler angles

- Inclination, the angle from the equatorial to the orbital plane (from $\hat{K}$ to $\hat{h}$)
- Right ascension of the ascending node (RAAN), angle between $\hat{I}$ and ascending node, which is the intersection of the orbital and equatorial planes, on the side where the satellite goes from the southern to northern hemispheres
- Argument of perigee: angle from the ascending node to eccentricity vector

[Stopped sharing to show 3D elements, resume sharing was not successful]



Special cases: $e=0$ (circular), $i=0$ (equatorial)
"Singular orbital element set"

Classical or Kepler orbital element set

a
 e
[time dep]
 i
 Ω
 ω

→ For example, true anomaly
Time element
Position angle

Description of orbital state
Orbital state vector Cartesian space

$$\vec{r}, \dot{\vec{r}} \rightarrow \begin{bmatrix} r_x \\ r_y \\ r_z \end{bmatrix}$$

The orbit in time: period and mean anomaly

$$a = \left(\frac{2}{r} - \frac{v^2}{\mu} \right)^{-1}$$

$$v = \sqrt{\mu \left(\frac{2}{r} - \frac{1}{a} \right)}$$

For circular orbit $r=a$
$$v_{\text{circ}} = \sqrt{\frac{\mu}{a}}$$

Orbital period
Curtis section 2.5

Area swept out in a given time interval

$$A_{t_1, t_2} = \frac{1}{2} h(t_2 - t_1)$$

$$A_{t_1, t_2} = \frac{1}{2} h (t_2 - t_1)$$

$$A_{0, \varphi} = \frac{1}{2} h \varphi = \pi a b$$

orbital period

$$\varphi = \frac{2\pi ab}{h} = \frac{2\pi a^2 \sqrt{1-e^2}}{\sqrt{p\mu}}$$

$$p = \frac{h^2}{\mu} \Rightarrow h = \sqrt{p\mu}$$

$$= \frac{2\pi a^2 \sqrt{1-e^2}}{\sqrt{\mu} \sqrt{a(1-e^2)}}$$

$$= \frac{2\pi}{\sqrt{\mu}} a^{3/2} \rightarrow [T]$$

Curtis (2.83)

Kepler's third law

$$\frac{\varphi^2}{a^3} = \frac{(2\pi)^2}{\mu}$$

True anomaly is useful geometrically, but its time dependence is unknown

Mean anomaly M , will change linearly in time

Areal angle

$$M = \frac{A(t)}{\pi ab} \cdot 2\pi = \frac{2A(t)}{ab}$$

$$\dot{M} = n = \text{const.}$$

Mean motion n is a constant

$$M(t) = n(t - t_p)$$

$$2\pi = M(p) = n p \Rightarrow n = \frac{2\pi}{p}$$

The relationship between mean and true anomaly is complex

Variants of orbital elements

- 1) Handy variations of classical elements e.g. rp, ra, instead of a, e
- 2) Changing the time element/position angle
- 3) Nonsingular elements - circular elements, equinoctial
- 4) Canonical elements

Lecture 4
9/10/2026

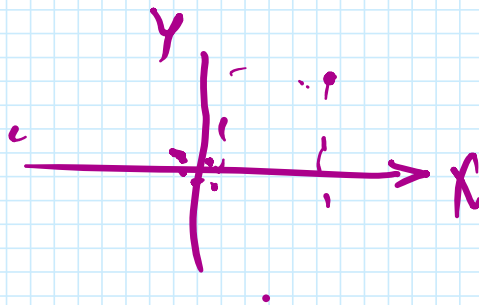
Nonsingular elements

Need:

- Circular orbits $e=0$, there's no perigee, so arg per and true anomaly undefined
- Equatorial orbits $i=0$, line of nodes does not exist, RAAN and argument of perigee undefined

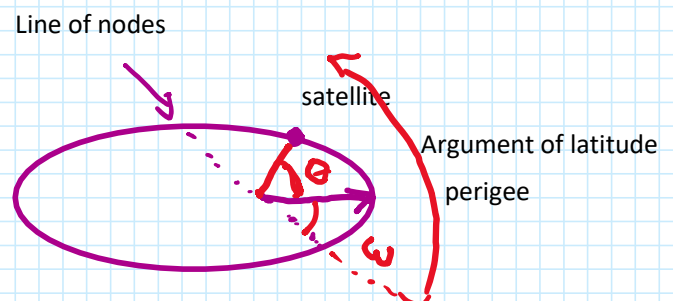
Circular and near circular orbits

- Very small eccentricity, imagine making a small change to the orbit



- Argument of latitude u
The angle from the ascending node to the satellite
If the orbit is not circular,

$$u = \omega + \theta$$



Equinoctial orbital elements (EqOE)

- Section 2.1 of Danielson, D. A., C. P. Sagovac, B. Neta, and L. W. Early. Semianalytic Satellite Theory. NPS-95-MA-002. Naval Postgraduate School, 1995. [DTIC ADA294557](#).
- Broucke, R. A., and P. J. Cefola. "On the Equinoctial Orbit Elements." Celestial Mechanics 5, no. 3 (1972): 303–10. [DOI 10.1007/BF01228432](#).
- Walker, M.J.H., Ireland, B. & Owens, J. A set modified equinoctial orbit elements. Celestial Mechanics 36, 409–419 (1985). [DOI 10.1007/BF01227493](#)
- Horwood, Joshua T., Nathan D. Aragon, and Aubrey B. Poore. "Gaussian Sum Filters for Space Surveillance: Theory and Simulations." Journal of Guidance, Control, and Dynamics 34, no. 6 (2011): 1839–51. [DOI 10.2514/1.53793](#).
- Baù, Giulio, Javier Hernando-Ayuso, and Claudio Bombardelli. "A Generalization of the Equinoctial Orbital Elements." Celestial Mechanics and Dynamical Astronomy 133, no. 11 (2021): 50. [DOI 10.1007/s10569-021-10049-1](#).

$$a = a$$

$$h = e \sin u$$

$$k = e \cos u$$

$$p = \tan \frac{i}{2} \sin \Omega$$

$$q = \tan \frac{i}{2} \cos \Omega$$

$$\lambda = M + \omega + \Omega$$

Mean longitude

- No large changes in these elements with small changes in orbit
- Mean longitude is like mean anomaly in that it increases linearly in time
- Computational efficiency

Canonical orbit elements

Hamiltonian mechanics - action-angle

- Actions = momenta
- Angles = coordinate

Delaunay elements

Delaunay elements

$$L = \sqrt{\mu a}$$

$$G = L \sqrt{1 - e^2} \quad \text{Total angular momentum, } h$$

$$H = G \cos i \quad \text{Polar component of angular momentum}$$

$$l = M \quad \text{Mean anomaly}$$

$$g = \omega \quad \text{Argument of perigee}$$

$$h = \Omega \quad \text{RAAN}$$

Poincaré elements

$$\mathcal{L} = L = \sqrt{\mu a}$$

$$\mathcal{l} = \Omega + \omega + M$$

$$g = \sqrt{2(L-G)} \cos(\omega + \Omega)$$

$$G = -\sqrt{2(L-G)} \sin(\omega + \Omega)$$

$$h = \sqrt{2(G-H)} \cos \Omega$$

$$\mathcal{H} = -\sqrt{2(G-H)} \sin \Omega$$



Delaunay

Like Delaunay elements, these are canonical conjugates

Like equinoctial elements, these are non-singular

Set	Position Variable	Singularities	Nature
Classical	M or θ	$e = 0, i = 0, i = 180^\circ$	Geometric
Equinoctial	λ	$i = 180^\circ$	Nonsingular
Modified Equinoctial	L (True Longitude)	None ($i = 180^\circ$ handled via I factor)	Universal
Delaunay	l (Mean Anomaly)	$e = 0, i = 0, i = 180^\circ$	Canonical
Poincaré	l (Mean Longitude)	None	Canonical / Nonsingular

Unbound orbits

Energy of the conic section

$\epsilon < 0$ bound orbits
 $\epsilon = 0$ parabolic
 $\epsilon > 0$ hyperbolic

$\epsilon = -\frac{\mu}{2a}$
 $\epsilon = \frac{\mu}{2a}$

Parabolic orbits

Curtis section 2.8

Ecc $e=1$, sma a not defined

$$r = \frac{p}{1 + \cos \theta}$$

$$r_p = \frac{p}{2} \leftarrow \text{semi latus rectum}$$

~~$$r_p = a(1-e)$$~~

$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} = 0$$

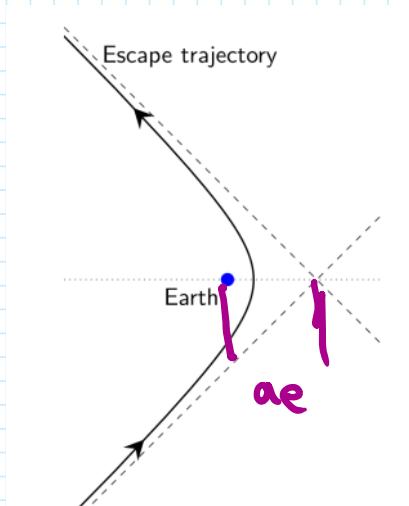
$$\epsilon = \frac{v^2}{2} - \frac{\mu}{r} = 0$$

$v_\infty = 0$ $r \rightarrow \infty$

$$= \frac{v_{esc}^2}{2} - \frac{\mu}{r} = 0$$

$$v_{esc} = \sqrt{\frac{2\mu}{r}}$$

Hyperbolic orbits
Curtis section 2.9



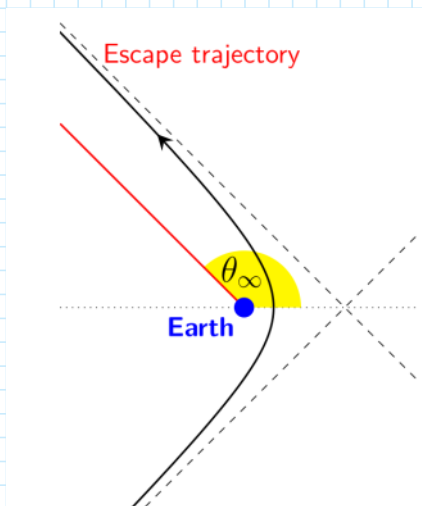
$$e > 1$$

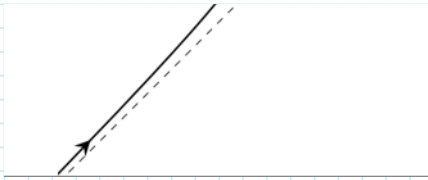
$$ae > a$$

$$\cos \theta = \frac{1}{e} \left(\frac{r}{a} - 1 \right)$$

$$\text{as } r \rightarrow \infty$$

$$\cos \theta_\infty = -\frac{1}{e}$$





Hyperbolic excess speed

$$\mathcal{E} = \frac{v^2}{2} - \frac{\mu}{r} = \frac{v_{\infty}^2}{2}$$

$$v_{\infty}^2 = v^2 - \frac{2\mu}{r} = v^2 - v_{esc}^2$$

Common earth orbits

Surface of earth $a=6378$ km, speed 7.9 km/s period 1 hour 24 minutes

Altitude of 500km, speed 7.6 km/s, period 1 hour 34 minutes "LEO"

At $a=42164$ km speed 3.1 km/s, period = 23 hours 56 minutes a few seconds; if $i=0$ "GEO"

Lecture 5
9/15/2026

Common orbits and their reasons

Criteria for choosing orbits

- Coverage of ground and/or visibility in space
- Lifetime of mission; low orbits brought in by atmospheric drag
- Cost of launch and/or maintenance; lower orbits are generally easier to reach
- Location of launch sites and the orbits that are accessible.
- Presence of forces that make it easier to maintain desired orbits, such as exposure to sun for charging solar panels

Useful orbits: divide into two categories

1. Orbit features driven by two-body dynamics
2. Orbit features driven by perturbations

Orbit Type	Key Orbital Parameters	Primary Use(s)
LEO (Low Earth Orbit)	Altitude < 1,000 km	Earth observation, space stations, crewed missions, high-resolution imaging
GEO (Geostationary / Geosynchronous Orbit)	Altitude ~35,786 km; orbital period matches Earth's rotation	Telecommunications, TV broadcasting, weather monitoring, continuous coverage over a fixed region
MEO (Medium Earth Orbit)	Altitude 1,000 km to 35,286 km	Navigation constellations (e.g., GPS, Galileo, GLONASS)
HEO (Highly Elliptical Orbit)	Eccentricity (e) > 0.5	High-latitude communication, prolonged coverage at apogee, specialized scientific missions

GEO = "Geosynchronous Equatorial Orbit"

Groundtracks

